EVER GUARDIAN documentation

motion & orientation:

Sudden impact detection :

When a person falls, there is a sudden and brutal acceleration in one or more directions. The accelerometer measures this variation an abnormally high acceleration peak compared to normal daily movements such as walking or sitting down

Rotation and orientation tracking :

Simultaneously, the body changes orientation radically from vertical to horizontal. The gyroscope measures this angular rotation. The combination of sudden acceleration and radical orientation change is what distinguishes a real fall from a fast normal movement like sitting down quickly or grabbing something

Post-event stillness confirmation :

After the impact, if the person is unconscious or unable to move, the sensor detects prolonged stillness. This is the final confirmation :sudden impact + orientation change + stillness = fall confirmed. Without this step, a simple quick gesture could trigger a false alarm

Fall detection logic :

The system processes these three signals in a sliding time window and applies either rule-based thresholds or a trained ML model to output a binary decision (fall or not fall)

Physiological Monitoring:

Heart Rate & HRV

Heart rate monitoring is one of the most critical features of this state tracker. The majority of elderly people today suffer from some form of heart disease whether diagnosed or not making continuous heart rate monitoring not just useful but essential. Beyond elderly-specific conditions, monitoring the heartbeat is fundamentally important for any person because the heart is the first organ to signal that something is wrong in the body. An abnormal heartbeat ;too fast, too slow, or irregular , is often the earliest detectable sign of a dangerous physiological event. HRV adds another layer by analyzing the variation between heartbeats, allowing tthis device to detect when the body is under stress, in pain, or approaching a cardiac event. This gives NANA' S GUARDIAN the ability to raise an alert before the person physically collapses

Blood Oxygen

When the blood carries insufficient oxygen, the brain is the first organ affected. Monitoring oxygen saturation continuously is critical because when blood oxygen level falls below a safe threshold, it leads to severe symptoms such as chest pain and fainting ,both of which directly cause falls and represent medical emergencies in elderly patients. By detecting a declining oxygen trend early, EVER GUARDIAN alerts the caregiver before these symptoms fully manifest. This is what makes this device predictive rather than just reactive ,it intervenes during the deterioration phase, not after the person has already collapsed

Skin Temperature

Skin temperature reveals the body's internal thermal state. Fever weakens the body and causes disorientation, increasing fall risk significantly. Hypothermia is a silent emergency that elderly people living alone are particularly vulnerable to. A sudden temperature drop can indicate a circulatory shock

glucose estimation :

Feature extraction: The sensor extracts three numerical values from the raw PPG signal(a signal which is extracted from photodetector of the sensor)

Light absorption ratio :how much light the blood absorbs at any given moment

Pulse travel time :how fast the blood is flowing through the vessels

Pulse shape :the precise geometry of each individual heartbeat

These three values are fed into a machine learning model and the model will maps them to an estimated glucose level

How Glucose Estimation Works :

The sensor onthis device emits light directly into the skin. This light travels through the skin tissue and reaches the blood vessels beneath. Inside the blood, glucose plays a direct role in changing the blood's viscosity ,meaning how thick or thin the blood is. When glucose levels rise, the blood becomes thicker. When glucose drops, the blood becomes thinner. This change in viscosity directly affects how the blood absorbs the light coming from the sensor. Thicker blood absorbs light differently than thinner blood, and the photodetector sitting next to the LED captures exactly this difference by measuring the light that bounces back through the skin. This returning light is converted into an electrical signal called the PPG signal , a continuous curve that rises and falls with every heartbeat. Because the blood viscosity changes with glucose, the shape of every pulse in this curve also changes ,how fast it rises, how fast it falls, how high it reaches, and how long the pressure wave takes to travel through the vessels. EVER GUARDIAN extracts these

shape characteristics from every heartbeat and feeds them into a machine learning model that was trained on thousands of real measurements where the PPG signal and the actual glucose level were recorded at the same time. The model learned exactly how glucose-driven viscosity changes affect the pulse shape. So whenthe sensor reads a new heartbeat, the model looks at the shape of that pulse and estimates the glucose level , all of this happening continuously, automatically, and without a single drop of blood.

sleep tracker:

Sleep can be detected automatically using built-in motion and heart-rate sensors. During the night, the system continuously monitors body movement and changes in heart rate patterns. When movement decreases and the heart rate becomes more stable, it recognizes that sleep has started. By analyzing variations in heart rhythm and subtle body movements, it can distinguish between light sleep, deep (profound) sleep, and brief awakenings. If movement increases or heart rate rises significantly, it identifies that the person is awake. All this data is analyzed through algorithms that classify sleep stages and calculate total sleep duration, sleep quality, and the times of falling asleep and waking up

GPS & Location Tracking:

Elderly people especially those with Alzheimer tend to wander and leave their home without being aware of it. They can get lost, end up in dangerous areas, or be exposed to extreme weather conditions. GPS tracking gives our state tracker the ability to monitor the person's location continuously and alert the caregiver the moment something abnormal is detected. Every alert sent by the device whether triggered by a fall, a physiological emergency, or a boundary crossing automatically includes the person's exact real-time location, so the caregiver always knows exactly where to go

Alerts & Reminders:

safety button :

False Alarm Suppression (User Interface) Before a Level 1 alert is transmitted, the device initiates a haptic feedback (vibration) Sound for 15 seconds. The user can manually cancel the alert by pressing the "Safety Button," which logs the event as a "False Positive" and resets the detection loop.

Medication reminder:

EVER GUARDIAN sends a scheduled notification directly to the caregiver at the defined medication times, reminding them that it is time to give the medication or perform a medical procedure such as an injection. This ensures that the caregiver never misses a critical medication moment and can follow up immediately with the elderly person

Titre 2